



Indian Association for the Cultivation of Science
(Deemed to be University under the *de novo* category)
BS-MS Program

Mid-Semester Examination-2025 (Autumn Semester-I)

Subject: Mathematics III

Subject Code(s): MAT 2101

Full marks: 25

Time allotted: 2 hrs

Answer all questions.

1. In a classroom there are N students.
 - (a) What is the probability that at least two students have the same birthday?
 - (b) What is the minimum value of N which secures probability $\frac{1}{2}$ that at least two of have a common birthday?

[5]
2. A binary communication channel carries data as one of the two types of signals denoted by 0 and 1. Owing to a noise, a transmitted 0 is sometimes received as a 1 and a transmitted 1 is received as a 0. For a given channel, assume a probability of 0.94 that a transmitted 0 is correctly received as a 0 and a probability 0.91 that a transmitted 1 is received as a 1. Further assume a probability of 0.45 of transmitting a 0. If a signal is sent, determine:
 - (a) probability that a 1 is received;
 - (b) probability that a 0 is received;
 - (c) probability that a 1 was transmitted, given that a 1 was received;
 - (d) probability that a 0 was transmitted, given that a 0 was received;
 - (e) probability of an error.

[6]
3. (a) Let X be a geometric random variable with parameter p . Show that for any positive integers m, n

$$P(X > m + n \mid X > m) = P(X > n).$$

- (b) Let X be a random variable taking values in $\{1, 2, \dots\}$. Suppose that X satisfies

$$P(X > m + n \mid X > m) = P(X > n)$$

for any positive integers m, n . Show that X is a geometric random variable.

[5]

4. Let X be a random variable with distribution function F_X which is strictly increasing. Let $Y = F_X(X)$. Show that Y is uniformly distributed. [4]

5. Consider a quiz game where a person is given two questions and must decide which question to answer first. Question 1 will be answered correctly with probability 0.8, and the person will then receive as prize Rs. 1000, while question 2 will be answered correctly with probability 0.5, and the person will receive as prize Rs. 2000. If the first question attempted is answered incorrectly, the quiz terminates, i.e., the person is not allowed to attempt the second question. If the first question is answered correctly, the person is allowed to attempt the second question. Which question should be answered first to maximize the expected value of the total prize received? [5]